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Towards Robust Machine-Learned Interatomic Potentials for Radiation-Damaged Waste Immobilization Materials

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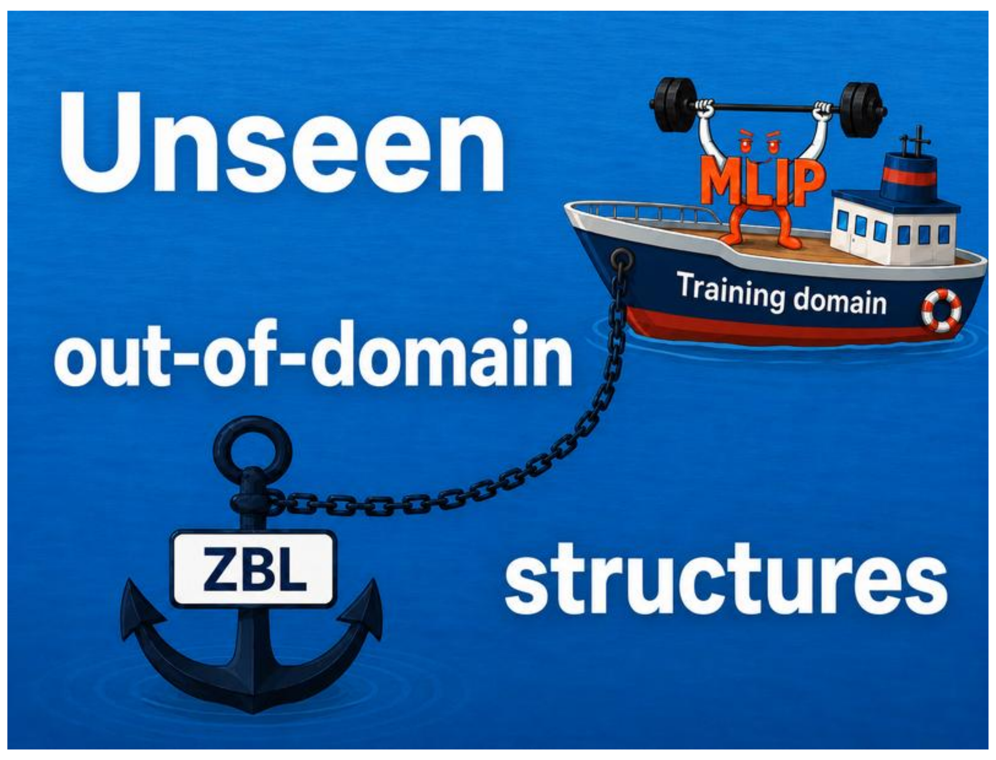
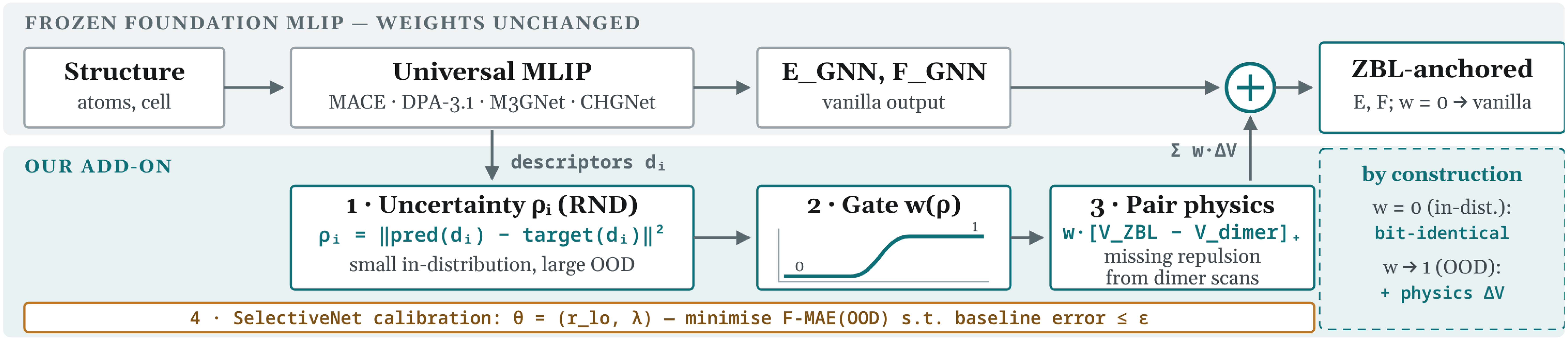
Nuclear waste forms accumulate strongly non-equilibrium local environments — recoil events, transmutation, defect build-up. Interatomic potentials must stay stable far from equilibrium yet accurate for chemically complex oxides: on our distorted perovskite-family set (19 elements, DFT/VASP) force R^2 is 0.65 (MACE-MH-0) / −0.03 (DPA-3.1), and fine-tuning destroys the inherited baseline. We add analytic short-range physics at inference time, only where the model is provably uncertain.



1.Method

Grey — frozen foundation model; teal — the add-on (steps 1–4).
Where the gate is silent (w = 0) the output is exactly vanilla — the baseline is preserved bit-for-bit.

$$E = E_{GNN} + \Sigma\, w(\rho)\cdot \Delta V(r) \quad \cdot \quad F = F_{GNN} - \lambda\, \Sigma\, w(\rho)\cdot \Delta V'(r)$$



2.The gating signal decides everything: RND novelty vs k-NN distance

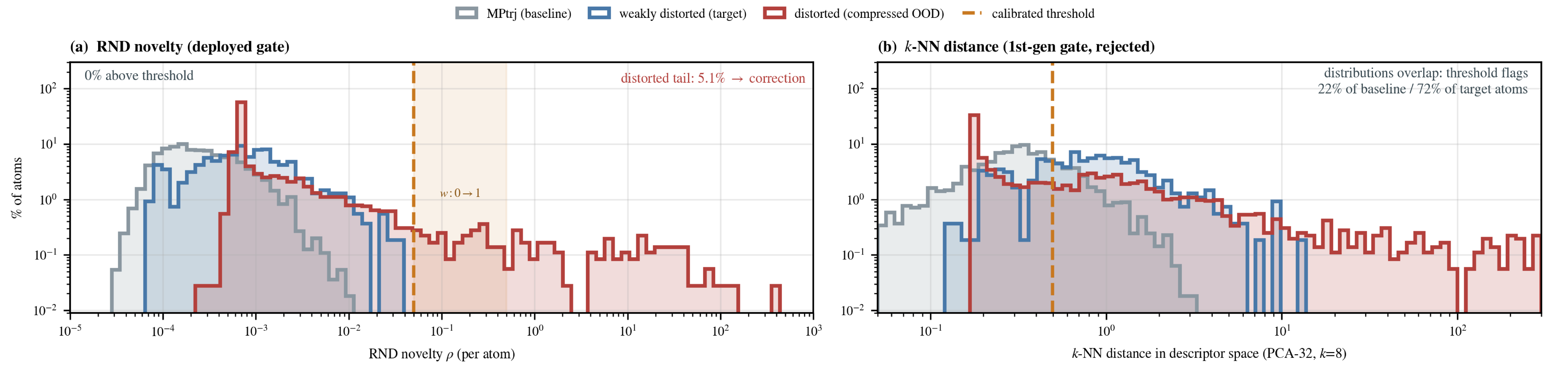


Fig. 1 | Both signals on the same atoms (11 099 baseline / 545 weakly distorted / 3 605 distorted).
(a) RND separates with zero errors (max baseline ρ = 0.010, weakly distorted 0.037, vs r_lo = 0.05); the distorted tail (5.1%, ρ up to 395) receives the correction; over the shaded band w ramps smoothly to keep forces continuous.
(b) The k-NN gate overlaps: p90 flags 22% of baseline / 72% of the weakly distorted set (force R^2 → −3.0 / −2.0) — rejected.

3.Accuracy on three held-out sets (force R^2, vanilla → anchor)

Tinted columns joined by ≡: anchor is bit-identical to vanilla there (gate silent → correction ≡ 0).

Model	MPtrj — baseline			weakly distorted — target			distorted (OOD)		distorted ΔF-MAE	overhead (fused)
	vanilla		*anchor*	*vanilla*		*anchor*	*vanilla*	*anchor*		
MACE-MH-0	0.986	≡	0.986	0.998	≡	0.998	0.650	0.829	-6.7%	×1.07
DPA-3.1	0.963	≡	0.963	0.995	≡	0.995	-0.033	0.733	-17.0%	×1.40
M3GNet	0.597	≡	0.597	0.992	≡	0.992	0.104	0.782	-17.1%	×1.17
CHGNet	0.936	≡	0.936	0.990	≡	0.990	0.005	0.014	-1.9%	×1.20

Descriptor dims: 256 / 128 / 64 / 64. CHGNet: the physics works (ungated ZBL 0.36) but its 64-d descriptors give no novelty tail.
Composition-disjoint split (MACE): 0.855 → 0.898. Energies excluded on the distorted set (inconsistent E0). Overhead on top of 12.9–50.1 ms vanilla E/F; physics ~2 ms; +1–4 MB.

4.Against fine-tuning

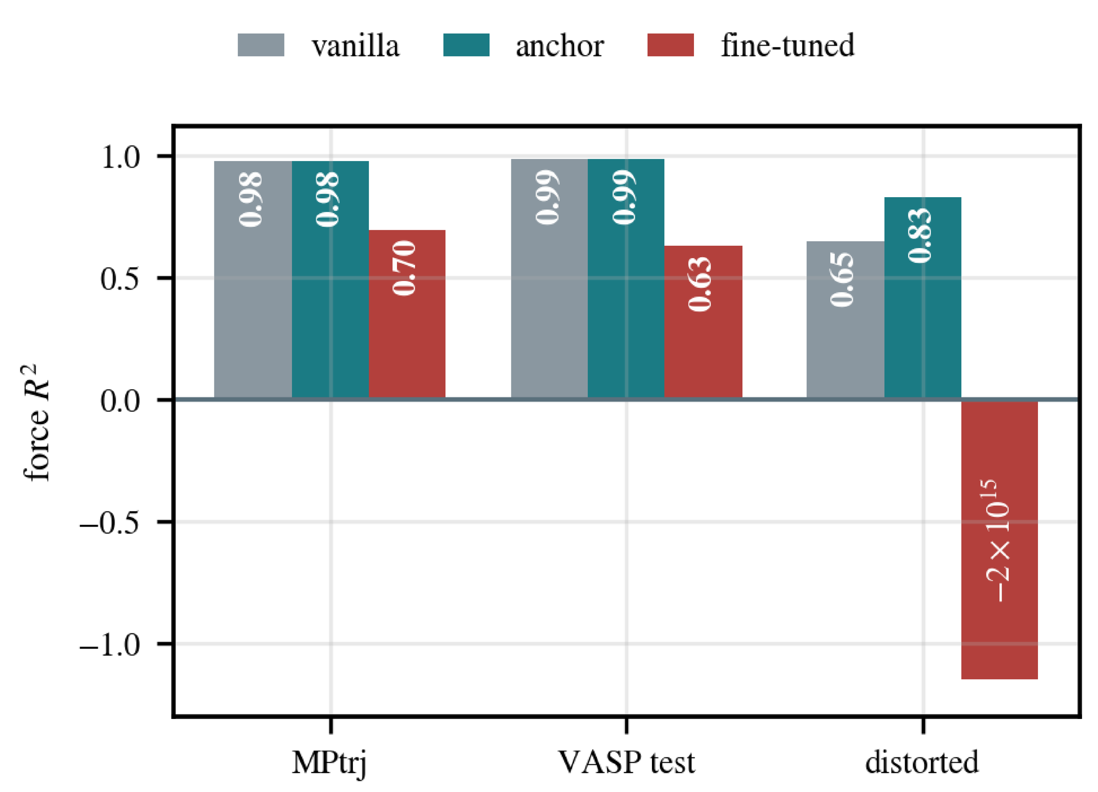


Fig. 2 | Fine-tuning degrades every in-distribution set (MPtrj 0.98 → 0.70, force MAE 0.044 → 0.145 eV/Å; VASP test 0.99 → 0.63), diverges on the distorted set and shrinks element coverage from 89 to 19 species; the anchor stays bit-identical wherever vanilla is good.

5.One recipe, four MLIPs

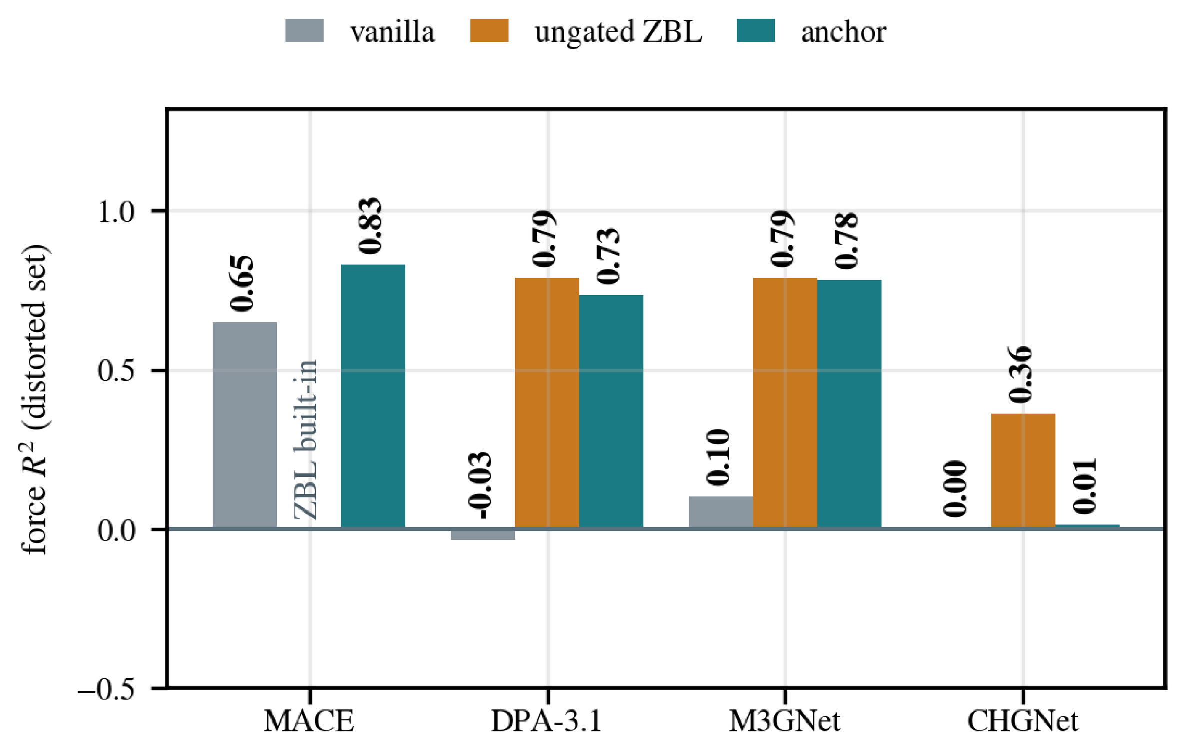


Fig. 3 | Identical pair-physics code on four MLIPs; only descriptors and thresholds are recalibrated. The same ungated ZBL, evaluated on the MPtrj baseline, collapses to force R^2 between −243 and −3609 — without the gate the physics destroys the baseline; the anchor keeps it bit-identical.

6.Compression stress test

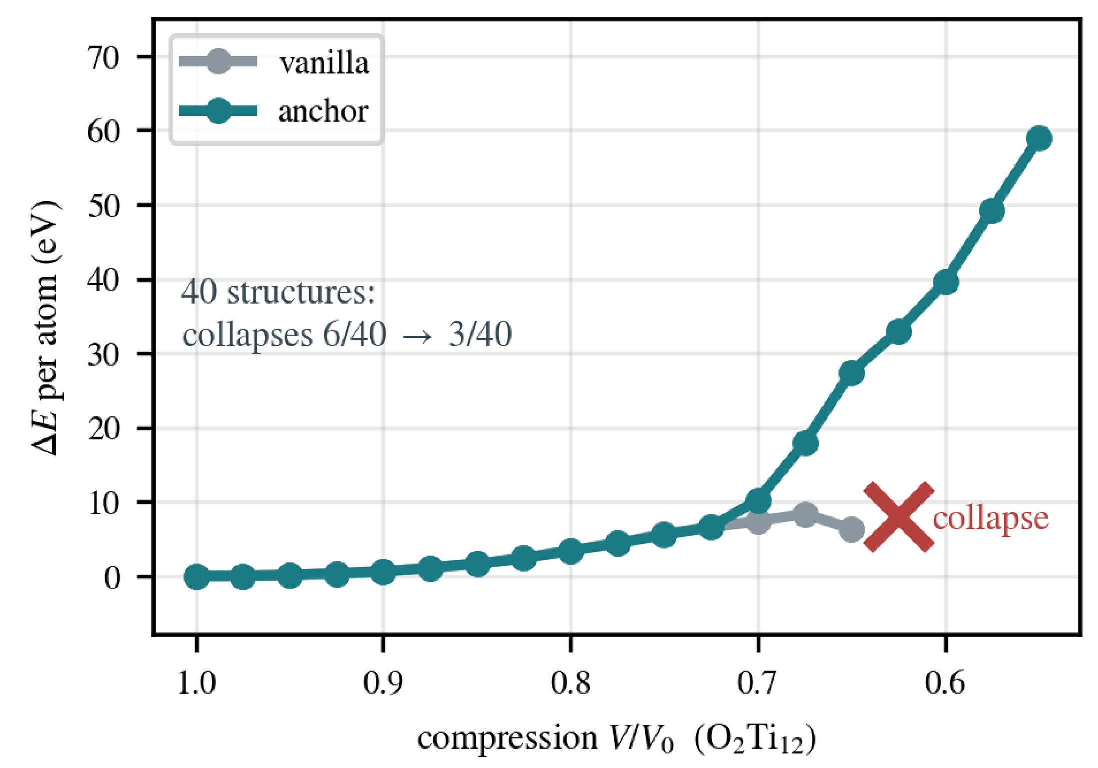


Fig. 4 | Isotropic compression with FIRE relaxation: vanilla collapses at V/V₀ = 0.625 (E → −946 eV/atom, d_min 0.22 Å), the anchored model stays smooth; over 40 structures collapses halve, never worse.

7.Conclusions

- ✓ OOD forces restored distorted-set force R^2 0.73–0.83 on three of four MLIPs, zero weight updates.
- ✓ Do-no-harm guaranteed baseline/target bit-identical — in statics and MD; elasticity, WBM, 2-MeV cascades unchanged.
- ✓ The signal is the method RND novelty separates (0% false positives); k-NN does not; ungated physics destroys the baseline.
- ✓ Deployment-ready +1–4 MB, ×1.07–1.4 cost, LAMMPS MD to 70 k atoms/GPU — groundwork for simulating radiation-induced disorder in waste forms.

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github.com/Mirual/ZBL_anchored

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